

ATIVIDADE DE APOIO – ENSINO MÉDIO

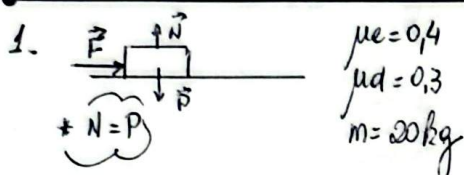
Componente Curricular: Física - Dinâmica

Professor (a): Rosana C. Pádua

Assunto (s): Força de Atrito I

Data: Fevereiro - 2026

Série: 2ª



$$\mu_e = 0,4$$

$$\mu_d = 0,3$$

$$m = 20 \text{ kg}$$

$$F_{at\text{em}} = \mu_e \cdot N$$

$$F_{at\text{em}} = 0,4 \cdot 200$$

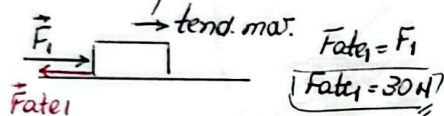
$$F_{at\text{em}} = 80 \text{ N}$$

$$F_{at\text{d}} = \mu_d \cdot N$$

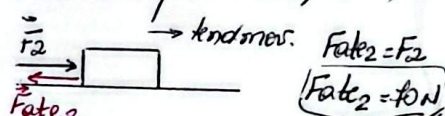
$$F_{at\text{d}} = 0,3 \cdot 200$$

$$F_{at\text{d}} = 60 \text{ N}$$

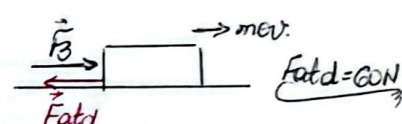
* $F_1 = 30 \text{ N}$ ($F_1 < F_{at\text{em}}$)
Bloco em repouso ($\vec{F}_R = \vec{0}$; $F_R = 0$)



* $F_2 = 10 \text{ N}$ ($F_2 < F_{at\text{em}}$)
Bloco em repouso ($\vec{F}_R = \vec{0}$; $F_R = 0$)



* $F_3 = 90 \text{ N}$ ($F_3 > F_{at\text{em}}$)
Bloco em movimento



2. $m = 10 \text{ kg}$
 $\mu_e = 0,3$
 $\mu_c = 0,25$
 $F = 80 \text{ N}$

$$F_{at\text{em}} = \mu_e \cdot N$$

$$F_{at\text{em}} = 0,3 \cdot 100$$

$$F_{at\text{em}} = 30 \text{ N}$$

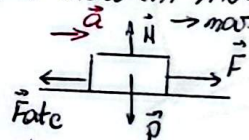
a) $F < F_{at\text{em}} \Rightarrow$ bloco em repouso ($\vec{F}_R = \vec{0}$; $F_R = 0$) $\therefore F_{ate} = 20 \text{ N}$

b) $F = 35 \text{ N}$ ($F > F_{at\text{em}}$) $\Rightarrow$ bloco em movimento

$$F_{at\text{c}} = \mu_c \cdot N$$

$$F_{at\text{c}} = 0,25 \cdot 100$$

$$F_{at\text{c}} = 25 \text{ N}$$

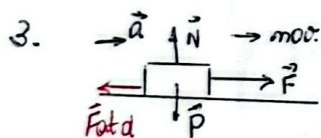


2ª Lei $\Rightarrow F_R = m \cdot a$

$$F - F_{at\text{c}} = m \cdot a$$

$$35 - 25 = 10 \cdot a$$

$$10 = 10 \cdot a \Rightarrow a = 1 \text{ m/s}^2$$



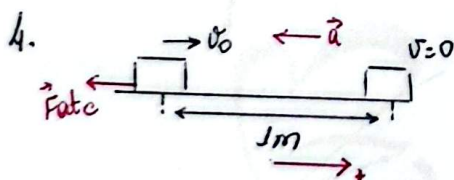
$$F = 100 \text{ N}; a = 1 \text{ m/s}^2; \mu = 0,1$$

$$F_R = m \cdot a \Rightarrow F - F_{at\text{d}} = m \cdot a$$

$$F - \mu_d \cdot N = m \cdot a \quad (N = P) \Rightarrow 100 - 0,1 \cdot 100 = m \cdot 1$$

$$100 - 10 = m \Rightarrow 2m = 100$$

$$m = \frac{100}{2} \Rightarrow m = 50 \text{ kg}$$



$$v_0 = 2 \text{ m/s}$$

$$\Delta S = 1 \text{ m}$$

$$\mu_c = ?$$

$$v^2 = v_0^2 + 2 \cdot a \cdot \Delta S$$

$$0 = 2^2 + 2 \cdot a \cdot 1$$

$$-2a = 4$$

$$a = -\frac{4}{2} \Rightarrow a = -2 \text{ m/s}^2$$

2ª Lei $\Rightarrow F_R = m \cdot a$

$$F_{at\text{c}} = m \cdot a$$

$$\mu_c \cdot N = m \cdot a$$

$$\mu_c \cdot mg = m \cdot a$$

$$\mu_c \cdot g = a$$

$$\mu_c \cdot 10 = 2$$

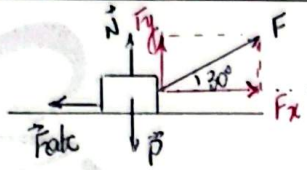
$$\mu_c = 0,2$$

Cuidado!!

• Equação de Torricelli é uma fórmula algébrica
 v_0, v, a têm sinal (lado eixo)

• 2ª Lei de Newton: é fórmula modular
 F_R, a entram + na fórmula, sempre.

5. $F = \frac{P}{2}$
 $\mu_c = ?$



$$F_x = F \cdot \cos 30^\circ$$

$$F_x = \frac{P}{2} \cdot \frac{\sqrt{3}}{2}$$

$$F_x = \frac{\sqrt{3} \cdot P}{4}$$

$$F_y = F \cdot \sin 30^\circ$$

$$F_y = \frac{P}{2} \cdot \frac{1}{2}$$

$$F_y = \frac{P}{4}$$

* Vertical: bloco em repouso ($\vec{F}_R = \vec{0}$; $F_R = 0$)

$$N + F_y = P$$

$$N + \frac{P}{4} = P \Rightarrow N = P - \frac{P}{4} \Rightarrow N = \frac{3P}{4}$$

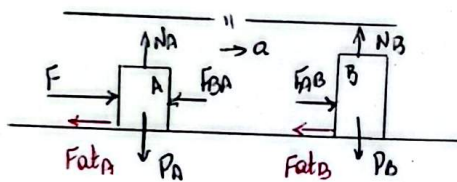
* Horizontal: vcke ($\vec{F}_R = \vec{0}$; $F_R = 0$)

$$F_x = F_{atc} \Rightarrow F_x = \mu_c \cdot N$$

$$\frac{\sqrt{3} \cdot P}{4} = \mu_c \cdot \frac{3P}{4}$$

$$\mu_c = \frac{\sqrt{3}}{3} \therefore \mu_c = \frac{1}{\sqrt{3}}$$

6. $F = 100\text{ N}$; $m_A = 6\text{ kg}$
 $\mu_c = \frac{1}{6}$; $m_B = 9\text{ kg}$
 $a = ?$



$$F_{atA} = \mu \cdot N_A$$

$$F_{atA} = \frac{1}{6} \cdot 60$$

$$F_{atA} = 10\text{ N}$$

$$F_{atB} = \mu \cdot N_B$$

$$F_{atB} = \frac{1}{6} \cdot 90$$

$$F_{atB} = 15\text{ N}$$

$$F_R = m \cdot a \Rightarrow A \Rightarrow F - F_{BA} - F_{atA} = m_A \cdot a$$

$$B \Rightarrow F_{AB} - F_{atB} = m_B \cdot a$$

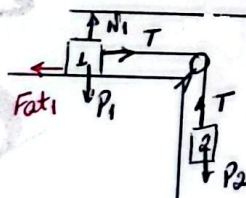
$$F - F_{atA} - F_{atB} = (m_A + m_B) \cdot a$$

$$100 - 10 - 15 = (6 + 9) \cdot a$$

$$75 = 15 \cdot a$$

$$a = \frac{75}{15} \Rightarrow a = 5\text{ m/s}^2$$

7. $m_2 = 2\text{ kg}$
 $v = 4,5\text{ m/s}$ (cke)
 $\mu = 0,4$



$$F_{at1} = \mu \cdot N_1$$

$$F_{at1} = 0,4 \cdot 10\text{ m}_1$$

$$F_{at1} = 4\text{ m}_1$$

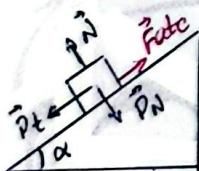
* vcke $\Rightarrow \vec{F}_R = \vec{0}$ ($F_R = 0$)

$$2 \Rightarrow T = P_2 \Rightarrow T = 20\text{ N}$$

$$1 \Rightarrow F_{at1} = T \Rightarrow 4m_1 = 20$$

$$m_1 = \frac{20}{4} \Rightarrow m_1 = 5\text{ kg}$$

8. $m = 20\text{ kg}$
 $\mu_c = 0,15$



$$N = P_N$$

$$N = P \cdot \cos \alpha$$

$$N = 200 \cdot 0,6$$

$$N = 120\text{ N}$$

$$F_{atc} = \mu_c \cdot N$$

$$F_{atc} = 0,15 \cdot 120$$

$$F_{atc} = 18\text{ N}$$

* $F_R = m \cdot a$

$$P_t - F_{atc} = m \cdot a$$

$$160 - 18 = 20 \cdot a$$

$$142 = 20 \cdot a$$

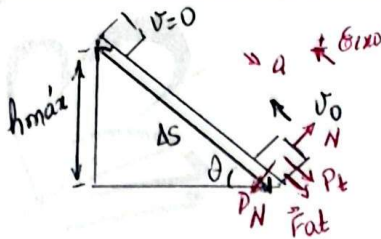
$$a = \frac{142}{20} \Rightarrow a = 7,1\text{ m/s}^2$$

$$P_t = P \cdot \sin \alpha$$

$$P_t = 200 \cdot 0,8$$

$$P_t = 160\text{ N}$$

9. $v_0 = 20 \text{ m/s}$
 $\mu = 0,25$
 $h_{\text{max}} = ?$



$$\begin{aligned} N &= P_N \\ N &= P \cos \theta \\ N &= 10 \text{ m} \cdot 0,8 \\ |N| &= 8 \text{ m} \end{aligned}$$

$$\begin{aligned} P_t &= P \sin \theta \\ P_t &= 10 \text{ m} \cdot 0,6 \\ P_t &= 6 \text{ m} \end{aligned}$$

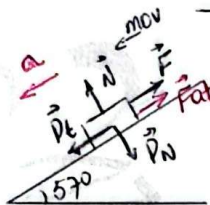
$$\begin{aligned} F_{at} &= \mu \cdot N \quad (N = P_N) \\ F_{at} &= 0,25 \cdot 8 \text{ m} \\ F_{at} &= 2 \text{ m} \end{aligned}$$

$$\begin{aligned} \Sigma F_x &= m \cdot a \\ P_t + F_{at} &= m \cdot a \\ 6 \text{ m} + 2 \text{ m} &= m \cdot a \\ 8 \text{ m} &= m \cdot a \\ |a| &= 8 \text{ m/s}^2 \end{aligned}$$

$$\begin{aligned} v^2 &= v_0^2 + 2 a \Delta S \\ 0 &= 20^2 + 2 \cdot (-8) \cdot \Delta S \\ 16 \Delta S &= 400 \\ \Delta S &= \frac{400}{16} \Rightarrow \Delta S = 25 \text{ m} \end{aligned}$$

$$\begin{aligned} \sin \theta &= \frac{h_{\text{max}}}{\Delta S} \\ 0,6 &= \frac{h_{\text{max}}}{25} \\ h_{\text{max}} &= 15 \text{ m} \end{aligned}$$

10. $m = 20 \text{ kg}$
 $\mu = 0,2$
 $F = 100 \text{ N}$



$$\begin{aligned} F_{at} &= \mu \cdot N \quad (N = P_N) \\ F_{at} &= 0,2 \cdot 200 \cdot 0,6 \\ F_{at} &= 24 \text{ N} \end{aligned}$$

$$\begin{aligned} P_t &= P \sin 37^\circ \\ P_t &= 200 \cdot 0,6 \\ P_t &= 120 \text{ N} \end{aligned}$$

$$\begin{aligned} \Sigma F_x &= m \cdot a \Rightarrow P_t - F - F_{at} = m \cdot a \\ 120 - 100 - 24 &= 20 \cdot a \\ 36 &= 20 \cdot a \\ a &= \frac{36}{20} \Rightarrow |a| = 1,8 \text{ m/s}^2 \end{aligned}$$